LIHAN ZHA

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Research Interests: Robotics, Machine Learning, Embodied Intelligence

EDUCATION

Princeton University Jul. 2024 – Jun. 2029 (Expected)

Ph.D. in Robotics

Tsinghua University Sep. 2020 – Jun. 2024

B.S. in Mathematics and Physic, B.E. in Energy and Power Engineering

Major GPA: 3.95/4.00 (Top 2/44) **GPA:** 3.84/4.00

Core Courses: Modern Control Theory (97); Statistical Inference (98); Fundamentals of Control Engineering (98); Fundamentals of Physics (93); Engineering Mechanics (100); Calculus (93); Machine Learning (100); Linear Algebra (88); Advanced Topics in Intelligent Robot (100); Deep Reinforcement Learning (98); Computer Vision (97); Electrical Engineering and Electronics (92); Mathematical Modeling and Data Analysis (100); Numerical Analysis (93); Engineering Thermodynamics (98)

Stanford University Summer Research Intern Jun. 2023 – Present

Massachusetts Institute of Technology Research Collaborator Mar. 2023 - Present

PUBLICATIONS

(*: Equal contribution. Listed alphabetically.)

1. **Lihan Zha,** Yuchen Cui, Li-Heng Lin, Minae Kwon, Montserrat Gonzalez Arenas, Andy Zeng, Fei Xia, Dorsa Sadigh, *Distilling and Retrieving Generalizable Knowledge for Robot Manipulation via Language Corrections*. ICRA, 2024. [arXiv]
2. Yanjiang Guo*, Yen-Jen Wang*, **Lihan Zha***, Zheyuan Jiang, Jianyu Chen, *DoReMi: Grounding Language Model by Detecting and Recovering from Plan-Execution Misalignment.* IROS, 2024. [arXiv]
3. **Lihan Zha**, Dongxiang Jiang, *An Ultra-Short Term and Short-Term Wind Power Forecasting Approach based on Optimized Artificial Neural Network with Time Series Reconstruction*. SPIES 2022. **Best presentation.** [Paper]

RESEARCH EXPERIENCES

Stanford's Intelligent and Interactive Autonomous Systems Group (ILIAD)

Advisor: Dorsa Sadigh, Assistant Professor at the Computer Science Department, Stanford University

Topic: Distilling and Retrieving Generalizable Knowledge for Robot Manipulation via Corrections Jun. 2023 – Present

The goal is to propose a novel method that enables the robot to respond to online language corrections and learn from these corrections so that the robot can reduce its error rate in future tasks.

- Designed a novel method, Distillation and Retrieval of Online Corrections (DROC), that effectively leverages a large language model (LLM) to directly infer how-to-respond, what-to-remember, and what-to-retrieve from human language corrections. Specifically, an LLM is prompted to first reason about relevant context required to respond to online corrections further identifying if the correction is about high-level task plans or low-level skill primitives. Then language feedback is distilled into re-usable knowledge via LLMs to improve generalization of both high-level plans and low-level skill primitives to new settings while reducing the number of human corrections needed over time. Finally, visual similarities of objects are leveraged for knowledge retrieval when language alone is not sufficient.
- Experiments across multiple long-horizon manipulation tasks show that DROC excels at (i) responding to online corrections, and (ii) adapting to new objects and configurations while consistently reducing the number of human corrections needed over time. DROC also achieves higher success rates and requires fewer corrections in new tasks compared to baselines such as Code as Policies (CaP) and variants with simpler implementation of distillation or retrieval mechanisms.

Intelligent Systems and Robotics (ISR) Lab, Tsinghua University

Advisor: Jianyu Chen, Assistant Professor at the Institute for Interdisciplinary Information Sciences, Tsinghua University

Topic: Grounding Language Model by Detecting and Recovering from Plan-Execution Misalignment Jan. 2023 – Jun. 2023

The goal is to propose a generic framework that enables execution-level high-frequency visual feedback and addresses the misalignment between the high-level plan and the low-level skill execution of embodied agents.

- Designed a framework that leverage large language models (LLMs) to generate the textual plan and the constraints for each skill in the plan for embodied agents. During the execution of each skill, the current observation of the agent and the constraint are fed to a vision language model (VLM) to determine whether the constraint is violated, which is framed as a binary visual question answering (VQA) task. If some constraints are found to be violated, which indicates that the plan and execution may be misaligned, the language model is immediately called to re-plan for timely recovery.

- Fine-tuned the VLM on a self-collected dataset to further increase the accuracy of constraint detection, and showed that the fine-tuning yields benefits for unseen new tasks. Trained the low-level locomotion policy of the humanoid via reinforcement learning, and designed model-based methods for manipulation policies.
- Experiments showed our proposed method results in high task success rate and low execution time in both humanoid and manipulator tasks compared to state-of-the-art baselines, e.g., Inner Monologue. We also demonstrated the robustness of our method under large environmental perturbations and the quality of LLM-generated constraints through ablation study.

Computational Cognitive Science (Cocosci) Lab, Massachusetts Institute of Technology

Advisor: Josh Tenenbaum, Professor at the Department of Brain and Cognitive Sciences, MIT

Topic: Online Learning and Exploration with Energy-Based Diffusion Models Feb. 2023 – Present

The goal is to design an online learning algorithm that can solve long-horizon tasks with sparse reward from scratch.

- Proposed to use energy-based diffusion models as the backbone of the robot policy, with which the robot can effectively explore and learn long-horizon tasks from scratch through interacting with the environment. The policy generates a predicted state sequence conditioned on the current state and the goal state, and then uses a learned inverse dynamics model to get the corresponding actions. At the same time, we showed that the energy function, which is the integral of the noise gradient field, can serve as an abstract density function and can guide the sampling process of the diffusion model to aid exploration.
- Experiments showed that the proposed method significantly outperforms baseline methods, e.g., PPO and SAC, in challenging long-horizon sparse-reward tasks like Maze2D.

Department of Engineering Mechanics, Tsinghua University

Advisor: Yue Shao, Associate Professor (tenured) at the Department of Engineering Mechanics, Tsinghua University

Topic: Defect Tolerance of Bionic Spider Web Structures Dec. 2021 – Jun. 2022

The goal is to study the mechanical properties of bionic spider webs with defects under different loading conditions and derive a dimensionless formula between defect tolerance and defect properties.

- Applied finite element method (FEM) to study how bionic spider web structures react to defects mechanically based on simulation analysis utilizing ABAQUS. Studied complicated relationships between 12 parameters of defect and structure's mechanical tolerance, including defect type, defect size, defect location, maximum load, maximum displacement, etc. Designed small scale study and proposed reasonable hypothesis to reduce parameters complexity.
- Developed Python-ABAQUS script interface to automatically perform the whole process of simulation, including defect modeling, parameters import, job submission, output export and processing.
- Experiments showed that in far field, bionic spider web structures' defect tolerance of radially extending defects and circumferentially extending defects changes linearly with defect size, and gradient distributed material shows better defect tolerance than uniformly distributed material.

ACEDAMIC SERVICE

Reviewer, IROS 2024, CoRL Workshop 2023

SELECTED AWARDS

2024 Outstanding Graduate of Beijing, China (**Highest honor** for graduates in Beijing, China)
2023 National Scholarship (Top 3/329, **highest honor** in the department)
2022 Jiang Nan Xiang Scholarship (Top 2/329, **highest honor** awarded to juniors in the department)
2022 Scholarship for Innovation in Science and Technology (Top 10/329)
2022 **First prize** in Excellent Student Research Training Contest (Top **20/3200** projects)
2021 Chengzhi - Excellence Scholarship (Top 4/167, **highest honor** for sophomores in Department of Energy and Power Engineering)
2021 Scholarship for Comprehensive Excellence (Top 15/330)
2021 Scholarship for Social Work Excellence (Top 10/330)

TECHNICAL SKILLS

Programming Language	Python, C/C++, HTML/CSS, SQL, MATLAB
Tool and Software	PyTorch, ROS, Git, Linux, Polymetis, AutoCAD, SolidWorks, ABAQUS, Fluent